

With the above command, you'll have a NAT instance number 123, which will send all traffic through a.b.c.d (see the documentation in the IPFW manual page for *same_ports and unreg_only*). This is just the NAT instance configuration; we haven't yet set up the rules for allowing traffic through the jails to the Internet. This part is a bit complex to understand and it took me many days to get this working. So I strongly recommend that you try this out in a virtual machine on your laptop/desktop first before trying it out on a machine to which you are connected over ssh.

I use a stateful firewall for outgoing traffic, so the first rule after permitting local traffic would be as follows:

```
ipfw add 200 check-state
```

But we need to add the incoming NAT rule *before* this one, so that the firewall can forward the NAT traffic correctly – i.e., if some connection was initiated from inside the jail, the packets related to it should go back there and not somewhere else where they don't belong.

```
ipfw add 199 nat 123 ip from any to a.b.c.d in
```

The above rule number 199 comes *before* our check-state rule number 200, so IPFW will check for NAT traffic arriving on public IP address a.b.c.d before checking for dynamic rules.

Let's suppose you have given jails IPs in the 10.0.0.0/24 subnet. Now, when you want to provide outgoing Internet access via NAT, you need to add rules like these before the rules for filtering outgoing traffic on the host. Now, let's consider a stateful outgoing traffic filtering rule on port 80 (i.e., let the machine access other http servers):

```
ipfw add 400 allow tcp from me to any 80 out setup keep-state
```

What this rule does is that whenever you initiate a connection to an external machine on port 80, it will first allow the SYN packet (setup flag) to pass, and then insert a dynamic rule for allowing incoming traffic appearing from port 80 on that IP address to your machine (the keep-state flag).

Before this rule, you need to add a rule so that traffic coming from 10.0.0.0/24 gets NATed and doesn't pass out directly. So the two rules (including the previous one) would be (the order is critical!) as shown below:

```
ipfw add 400 skipto 700 tcp from 10.0.0.0/24 to any 80 out
setup keep-state
ipfw add 400 allow tcp from me to any 80 out setup keep-state
```

IPFW allows the same index numbers for different rules, and when that happens, rules are applied sequentially as they were added. With the 'skipto 700' action in the above rule, we tell IPFW to jump to rule number 700 for all packets arriving from 10.0.0.0/24 and trying to leave the machine to port 80. We need the following rule as rule number 700:

```
ipfw add 700 nat 123 ip4 from 10.0.0.0/24 to not 10.0.0.0/24
out
```

The above rule will apply the NAT for all IPv4 addresses. NAT is not supported on IPv6 and the keyword ip means both v4 and v6.

Resource control

Jails can be resource limited using FreeBSD's Resource Accounting (RACCT). For this to work, you need to recompile the kernel with the following options:

```
options RACCT
options RCTL
```

See the FreeBSD documentation for kernel compilation – it's really easy. A couple of parameters are available that can be controlled using RCTL, which can be found in the man pages. I'll describe one of these parameters – *vmemoryuse*. It restricts the address space of a jail, effectively limiting its maximum memory usage.

```
rctl -a jail:test:vmemoryuse:deny=512M
```

The above command will add a rule to restrict the maximum address space to 512 megabytes. 

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